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A laser with instability reaching 4×10^{-17} based on a 10-cm-long silicon cavity at sub-5-K temperatures

Zhi-Ang Chen^{a,b}, Hao-Ran Zeng^{a,b}, Wen-Wei Wang^{a,b}, Han Zhang^{a,b,†}, Run-Qi Lei^{a,b}, Jian-Zhang Li^c, Cai-Yin Pang^c, She-Song Huang^d, Xibo Zhang^{a,b,e,f,*}

^a International Center for Quantum Materials, School of Physics, Peking University, Beijing 100871, China

^b Collaborative Innovation Center of Quantum Matter, Beijing 100871, China

^c Beijing Xinnan Zhike Optoelectronic Technology Co., Ltd, Beijing 100023, China

^d Beijing Physike Technology Co., Ltd, Beijing 100085, China

^e Hefei National Laboratory, Hefei 230088, China

^f Beijing Academy of Quantum Information Sciences, Beijing 100193, China

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ABSTRACT

The realization of ultra-stable lasers with 10^{-17} -level frequency stability has enabled a wide range of researches on precision metrology and fundamental science, where cryogenic single-crystalline cavities constitute the heart of such ultra-stable lasers. For further improvements in stability, increasing the cavity length at few-kelvin temperatures provides a promising alternative to utilizing relatively short cavities with novel coating, but has yet to be demonstrated with state-of-the-art stability. Here we report on the realization of a relatively long ultra-stable silicon cavity with a length of 10 cm and sub-5-K operating temperatures. We devise a dynamical protocol of cool-quiet quench measurement that reveals the inherent 10^{-17} -level frequency instability of the silicon cavity despite the substantially larger frequency noise induced by the cryostat vibration. We further develop a method for suppressing the cryostat-vibration-induced frequency noise under continuous cooling, and observe an average frequency instability of $4.3(2) \times 10^{-17}$ for averaging times of 4 to 12 s. Using the measured noise power spectral density, we compute a median linewidth of 9.6(3) mHz for the silicon cavity laser at 1397 nm, which is supported by an empirically determined linewidth of 5.7(3) mHz based on direct optical beat measurements. These results establish a new record for optical cavities within a closed-cycle cryocooler at sub-5-K temperatures and provide a prototypical system for using long cryogenic cavities to enhance frequency stabilities to the low- 10^{-17} or better level.

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1. Introduction

Advances in laser frequency stabilization have led to rapid progress in the studies of optical atomic clocks [1,2], precision metrology [3,4], and fundamental science [5–8]. In particular, highly stable optical local oscillators (namely ultra-stable lasers) are an indispensable ingredient of optical clock experiments that demonstrate excellent measurement precision with statistical frequency uncertainties on the order of 10^{-19} or better [7–10]. Recently, state-of-the-art ultra-stable lasers at wavelengths near 1542 nm have reached frequency stabilities in the middle of the 10^{-17} range,

which are fundamentally limited by the Brownian thermal noise of cryogenic single-crystalline silicon optical cavities with dielectric coatings [11–13]. Cryogenic silicon cavities with semiconductor crystalline coatings [14] hold promise for realizing an even lower level of Brownian thermal noise. However, reaching such a low thermal noise limit in the associated laser frequency stability has been encumbered by new noise sources like birefringent noise and other unclear residual noise [15,16]. Several alternative methods have also been studied, including engineering the cavity temperature [17,18], choosing different crystalline materials [19,20] and exploring new coating structures [21].

With a characteristic $\sqrt{T}/L$ dependence on the cavity length L and cavity temperature T , the fundamental Brownian thermal noise can be suppressed by increasing L and reducing T . This is straightforward to consider, but very challenging to realize at liquid-helium temperatures, which is due to the difficulty in

* Corresponding author.

E-mail address: xibo@pku.edu.cn (X. Zhang).

[†] Present address: Quantum Science Center of Guangdong-Hong Kong-Macao Greater Bay Area, Shenzhen 518045, China.

simultaneously achieving a relatively long cavity and effectively mitigating the increased frequency noise induced by the corresponding large-volume cryostat. Compared with the apparatus in Refs. [12,13,15], a new cryostat was recently designed and realized that reaches lower operating temperatures of about 3 K for a substantially longer silicon cavity with a length of 10 cm [22]. Progress on 1.7-K and sub-kelvin cryogenic silicon cavities has also been reported [23,24]. However, 10^{-17} -level laser frequency stability has yet to be demonstrated with relatively long cavities at few-kelvin temperatures.

In this article, we experimentally realize a 10-cm-long, sub-5-K silicon optical cavity with a median laser linewidth below 10 mHz at 1397 nm and a frequency instability of 4×10^{-17} . We first develop a dynamical experimental protocol for performing three-cornered-hat (TCH) measurements and demonstrate a 10^{-17} -level optical performance for the cryogenic silicon cavity even when the cryostat-vibration-induced frequency noise is at a substantially higher level. Under continuous cooling, we further identify two major uncontrolled noise sources that seriously hamper the stability of the silicon cavity, and accordingly demonstrate strategies that successfully overcome these obstacles. The resultant silicon cavity shows an optimized frequency instability of $4.3(2) \times 10^{-17}$ at averaging times of 4 to 12 s, which is close to the Brownian thermal noise floor of this cavity at 4.9 K (about 3.3×10^{-17} ; see the [Supplementary material](#)) and revealing better frequency stability at these averaging times than the previous best sub-5-K optical cavities [12,13,15]. Our work offers a promising platform for reaching laser stabilities in the low- 10^{-17} range or better based on relatively long silicon cavities at even lower temperatures.

2. Setup and methods

As shown in Fig. 1a, we designed and realized a 10-cm-long ultra-stable silicon optical cavity, referred to as Si1. The monocrystalline cavity spacer has an optical axis oriented along the crystalline (111) axis of silicon. A pair of high-reflectivity cavity mirrors, with radii of curvature of ∞ and 1 m respectively, are superpolished, coated with a dielectric coating technique, and subsequently optically contacted to the 9.79-cm-long spacer (calibrated under about 3 K), providing a finesse of about 3.6×10^5 . This cavity operates at sub-5-K temperatures in a two-stage Gifford–McMahon closed-cycle cryostat [22].

To prepare a clean and low-vibration few-kelvin cryogenic platform to accommodate the Si1 cavity, we implemented six technical improvements, denoted as improvements (1) to (6), with respect to our previous cryogenic apparatus [22] and an earlier version of a cryogenic silicon cavity (named Si0):

(1) To eliminate harmful outgassing processes that potentially contaminate the cavity mirrors, we thoroughly cleaned all in-vacuum surfaces, abandoned certain “standard practices” of the cryostat manufacturer Montana Instruments, and applied only high-vacuum-compatible material to maintain good thermal contact between adjacent surfaces (see the [Supplementary material](#)).

(2) To further suppress the condensation of gas molecules onto the cavity mirrors, we adopted a stringent vacuum criterion by reducing the in-sample-chamber pressure to the 10^{-7} -Torr high-vacuum regime before any cooling started (see the [Supplementary material](#)).

(3) To improve the rigidity of the mechanical structure supporting the silicon cavity vertically (red arrows in Fig. 1a inset), we

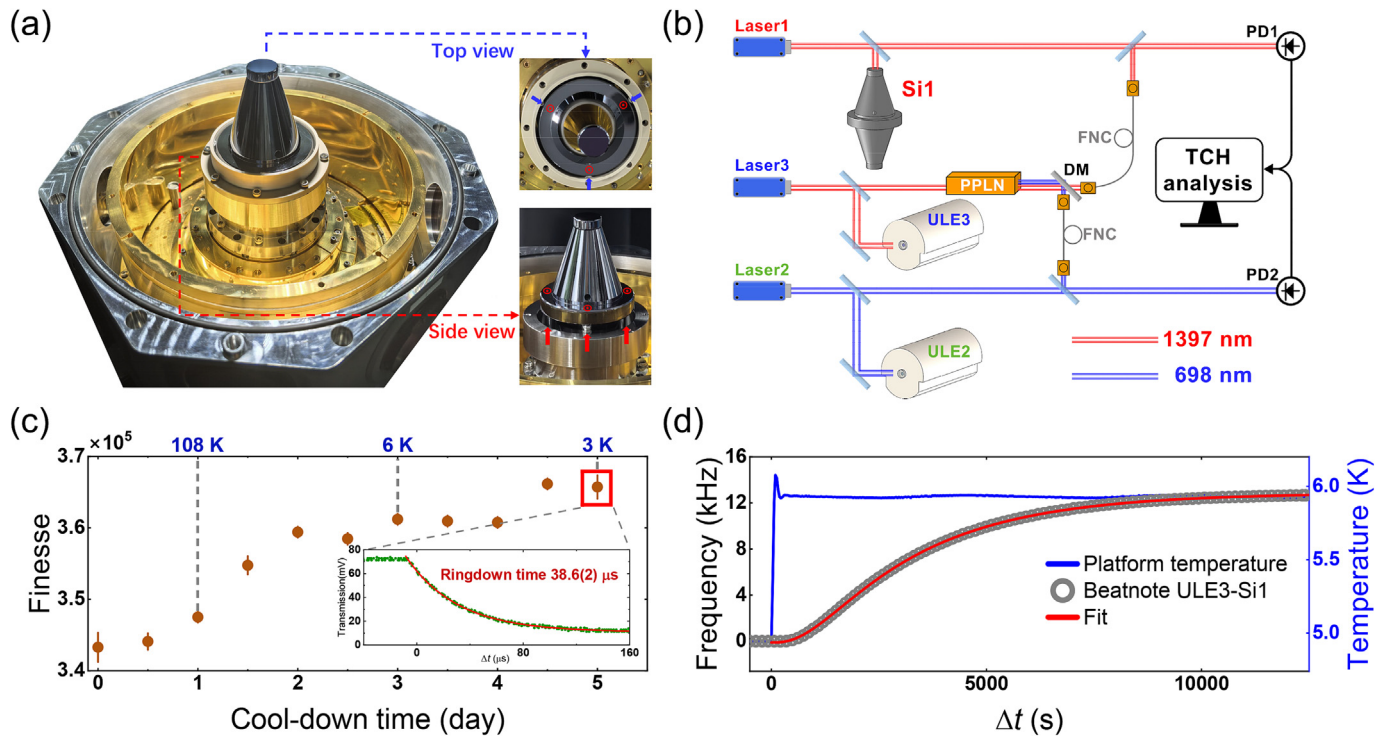


Fig. 1. Experimental setup and safe cooling down of a silicon cavity to few-kelvin temperatures with intact optical performance. (a) Silicon cavity (Si1) and its vacuum chamber for cryogenic cooling. In the insets, red arrows indicate a vertical support under the Si1 central ring at three points marked by centers of red circles, and blue arrows indicate a three-point fastening of the Si1 horizontal position (not shown in the side view). (b) Diagram of three-cornered-hat (TCH) frequency stability measurements based on the Si1 cavity and two room-temperature ultra-low-expansion (ULE) glass cavities, ULE2 and ULE3. DM: dichroic mirror; PD: photo-detector; FNC: fiber noise cancellation; PPLN: periodically poled lithium niobate waveguide for frequency doubling. (c) The Si1 cavity finesse (solid circles) remains stable and slightly increases during the cooling. Inset: a ringdown measurement for determining the finesse at a platform temperature of about 3 K. (d) Step-function response of the Si1 resonance frequency for *in situ* determination of cavity temperature, with improvements (1) to (4) implemented. The blue line is the platform temperature. Gray circles denote the beat frequency between Laser3 (based on the ULE3 cavity) and Laser1 (Si1). The red line shows a fit using a third-order low-pass filter model. Error bars represent 1σ statistical uncertainties.

built a semi-monolithic, all-stainless-steel, new supporting structure that has no shakable components within itself and realizes higher mechanical resonance frequencies (see the [Supplementary material](#)).

(4) To reduce the vibration transferred from cryostat coolers, we separated the baseplate of the sample chamber from that of the cryocooler chamber, and enabled an active vibration control platform (Table Stable, AVI-200-XL-LP), on which the sample chamber sat.

(5) To further suppress the frequency noise induced by cryostat cooler vibration, we designed and implemented a three-point fastening of the horizontal position of the Si1 cavity ([Fig. 1a](#), upper inset), with proper strain relief applied (see the [Supplementary material](#)).

(6) To protect the cavity against environmental fluctuations such as acoustic noise and air flow, we enclosed the entire optical path of the silicon cavity laser inside an acoustic shield made of sound-absorbing cotton.

[Table 1](#) lists a few configurations with various numbers of implemented improvements and experimental conditions.

We realized an ultra-stable laser (“LoongLaser-1”, or simply “Laser1”) by phase-locking it to the Si1 cavity based on the Pound-Drever-Hall (PDH) approach [25]. This laser (operated near 1396.8892 nm) was locked to the TEM₀₀ mode of Si1 with a 600-kHz locking bandwidth and controlled μ W-level incident optical power (see the [Supplementary material](#)). To measure frequency stabilities, we built two other stable lasers (“Laser2” and “Laser3”) that were independently phase-stabilized to two room-temperature, 30-cm ultra-low-expansion (ULE) glass cavities, referred to as ULE2 (operated near 698.4445 nm, close to the strontium clock transition) and ULE3 (near 1396.8900 nm) respectively; see [Fig. 1b](#). The three-laser system enabled us to measure optical beat frequencies (using K + K FXE frequency counters) and perform TCH frequency stability analysis to determine the frequency instability for each laser, represented by the modified Allan deviation [12,26]; see the [Supplementary material](#). Here, the frequencies of the 698-nm Laser2 and 1397-nm Laser3 were bridged by frequency doubling via a single-pass periodically poled lithium niobate (PPLN) waveguide [27]. All frequency synthesizers and counters were referenced to a 10-MHz rubidium clock (Chengdu Synchronization Technology, STM-Rb-N).

3. Results

3.1. Cooling the Si1 cavity without optical contamination

While a silicon cavity operating at several kelvins naturally serves as a powerful cryogenic pump, such pumping can potentially lead to optical contamination of the cavity mirrors. To protect these mirrors, we prepared a clean environment by implementing improvements (1) and (2), and calibrated the initial optical properties of the Si1 cavity under room temperature. For the TEM₀₀ mode, we performed ringdown measurements [28,29] under low and

high vacuum with pressures of $P_{LV} \approx 4$ Torr and $P_{HV} \approx 4 \times 10^{-7}$ Torr, and determined the effective finesse of the Si1 cavity to be $\mathcal{F}_{1,LV} = 1.68(10) \times 10^5$ and $\mathcal{F}_{1,HV} = 3.43(3) \times 10^5 > \mathcal{F}_{1,LV}$ at these pressures respectively, which is qualitatively consistent with the influence of water absorption near 1397 nm. The Si1 cavity finesse for the TEM₀₁ mode is similar to that for TEM₀₀ with a fractional difference of -3.5% , demonstrating good uniformity of mirror finesse.

The cleanness of the high-vacuum environment was enhanced during a five-day cooling process, which was evidenced by the robustness of the Si1 optical properties. During the cooling, the pressure decreased to about 1×10^{-8} Torr, and the coupling efficiency of Laser1 to the Si1 TEM₀₀ mode remained unchanged within the measurement uncertainty without reoptimization. As shown in [Fig. 1c](#), the measured Si1 cavity finesse slowly increased to a value 7% larger than $\mathcal{F}_{1,HV}$ under a cryostat platform temperature of about 3 K. Under this platform temperature, we further estimated a mean loss coefficient $L_M \approx 3$ ppm and a mean transmission coefficient $T_M \approx 5$ ppm $> L_M$ of the cavity mirrors for the TEM₀₀ mode (see the [Supplementary material](#)). Furthermore, even after one full thermal cycle from room temperature to several kelvins (with a four-month cryogenic operation period) and back to room temperature, the Si1 cavity finesse at high vacuum remained $\mathcal{F}'_{1,HV} = 3.42(3) \times 10^5$, showing almost no sign of degradation with respect to $\mathcal{F}_{1,HV}$. These observations support that our clean cryogenic environment has vastly suppressed potential optical contamination and kept the Si1 cavity optically intact at temperatures of a few kelvins. In sharp contrast, the earlier Si0 cavity showed subtle evidences of contamination, exhibiting substantially lower and spatially inhomogeneous finesse that further degraded after a few months (see the [Supplementary material](#)).

To quantitatively characterize the average *in situ* temperature of the whole Si1 cavity, we identified the cryostat platform temperature as a convenient approximate “readout port” for this goal based on the following considerations. First, we have previously measured the cavity temperature at its center ring and found it to be fairly close to the platform temperature (0.13 K above the latter) [22]. Here, this sensor was removed to eliminate the associated thermal conduction path, and we expected the cavity to be more efficiently cooled to a slightly lower temperature that is even closer to the platform temperature. Second, we determined the average *in situ* Si1 cavity temperature using a quench technique based on measurements of cavity resonance frequencies and knowledge of the silicon coefficient of thermal expansion (CTE) [30,31]. As shown in [Fig. 1d](#), with improvements (1) to (4) implemented, we suddenly changed the set platform temperature by 1 K from 4.94 to 5.94 K, and observed a smooth crossover of the Laser3-Laser1 beat frequency from one value to another. This crossover behavior can be fitted using a third-order low-pass filter [22] with time constants of 500, 500, and 2700 s, and a total frequency change of 12.7 kHz can be used to determine the cavity temperature. Specifically, we used a model of cryogenic material properties by the National Institute of Standards and Technology (<https://trc.nist.gov/cryogenics/calculators/propcalc.html>), which is supported by experiments [32–34]. This model shows that, for low temperatures ($T \leq 10$ K), the silicon CTE can be approximately described by a T^3 scaling, namely, $CTE_{Si} \approx \alpha T^3$, where α is a coefficient. Integrating this scaling with respect to T leads to a total frequency change of $\frac{3}{4}[(T_i + 1)^4 - T_i^4]$, with T_i being the initial cavity temperature. Based on a recent study on ultra-stable silicon cavities at several kelvins [30], where CTE_{Si} was measured to be $4.5 \times 10^{-11}/K$ at 4.9 K, we extracted $\alpha \approx 3.8 \times 10^{-13}/K^4$ and thus determined $T_i \approx 4.9$ K for the Si1 cavity, which was indeed close to the initial platform temperature of 4.94 K. In another quench measurement, we

Table 1
Experimental configurations of the silicon cavity.

Cavity configuration	Implemented improvements	Cavity temperature	Related figures
Si1a	(1) to (4)	4.9 K	Figs. 3, 1d
Si1b	(1) to (5)	4.9 K	Figs. 3, 4b
Si1c	(1) to (6)	4.9 K	Fig. 4
Si1c'	(1) to (6)	3.3 K	Fig. 4a inset
Si1-CQOM	(1) to (4)	near 3.3 K	Figs. 2, 3
Si0	none	3 K	Fig. 3

determined $T_i \approx 3.1$ K, again close to the corresponding initial platform temperature of 3.28 K. Taking these into account, we chose the platform temperature as a convenient characterization of the Si1 cavity temperature T in the sub-5-K regime.

3.2. Dynamically revealing the 10^{-17} -level frequency stability of a cryostat-vibration-free Si1 cavity by “Cool-Quiet” quench measurement

At sub-5-K temperatures, the Si1 cavity has a fundamental Brownian thermal noise floor at and below 3.3×10^{-17} , which is computed using $T = 4.9$ K, $L = 9.79$ cm, a 1397-nm wavelength and dielectric coating properties similar to those in Ref. [17]; see Fig. 2 and the Supplementary material. Under these conditions, the Si1 cavity is estimated to have a frequency drift rate below 1 mHz/s; see the Supplementary material. Because vibrational noise due to the cryostat can often overwhelm such a low Brownian thermal noise floor [23], it is desirable to develop a diagnostic method for demonstrating and verifying the high optical performance of such cavities without being restricted by cryostat vibration.

We developed a protocol of cool-quiet quench measurement (CQQM) to determine, in a manner insensitive to the cryostat vibration, the Laser1 frequency stability due to the fundamental optical performance of the Si1 cavity and precision of the associated electro-optical control system. In this protocol shown in Fig. 2a, the Si1 cavity was initially cooled to a low temperature in the range of a few kelvins (namely the “Cool” stage), where the whole cryostat vibration effect was contained in the Laser1 frequency noise. Then the cryostat cooling was suddenly disabled at time $t = 0$ (namely the “Quench” action), after which the compressor and other parts took some time to relax into a new equilibrium. When the new, quieter mechanical equilibrium was reached, we performed TCH measurements to extract the Laser1 frequency instability with the cryostat suspended and the corresponding vibration temporarily removed (namely the “Quiet” stage). Here, the measurement time was properly constrained such that the cavity temperature did not drift too much.

We performed CQQM with an initial platform temperature of 3.3 K and observed 10^{-17} -level frequency instability for the Si1 cavity with improvements (1) to (4) implemented. As shown in Fig. 2b, the Si1 frequency instability reached $9.9(7) \times 10^{-17}$ at an averaging time of $\tau = 1$ s, improved to $6.8(5) \times 10^{-17}$ at 2 s and $6.6(1.3) \times 10^{-17}$ at 6 s, and changed towards 1×10^{-16} at $7 \sim 10$ s. Such stability not only surpassed Si1’s own room-temperature stability, but also outperformed the reference cavities ULE2 and ULE3, which both showed frequency instabilities below 2×10^{-16} at $1 \sim 10$ s. The Si1 instability revealed by CQQM reached the same order of magnitude as the typical Brownian thermal noise floor at 4.9 K (red dashed line in Fig. 2b), which signifies that, except for the cryostat vibration, most noise sources have been well controlled to the 10^{-17} level. By contrast, neither the 10^{-17} -level CQQM result nor the 10^{-16} -level instability at room temperature has ever been observed for the Si0 cavity. In next sections, we chose a platform temperature of 4.9 K under a “slow” cooling mode with minimum cryostat vibration.

3.3. Overcoming two obstacles to achieving 10^{-17} -level frequency stability under continuous cooling

With continuous cooling of the cryostat, we identified two major obstacles as two types of uncontrolled technical perturbations that are difficult to model but severely degrade the frequency stability of a silicon cavity at a few kelvins. The first obstacle lies in uncontrolled optical scattering due to mirror contamination. The second stems from uncontrolled vibration amplified by non-

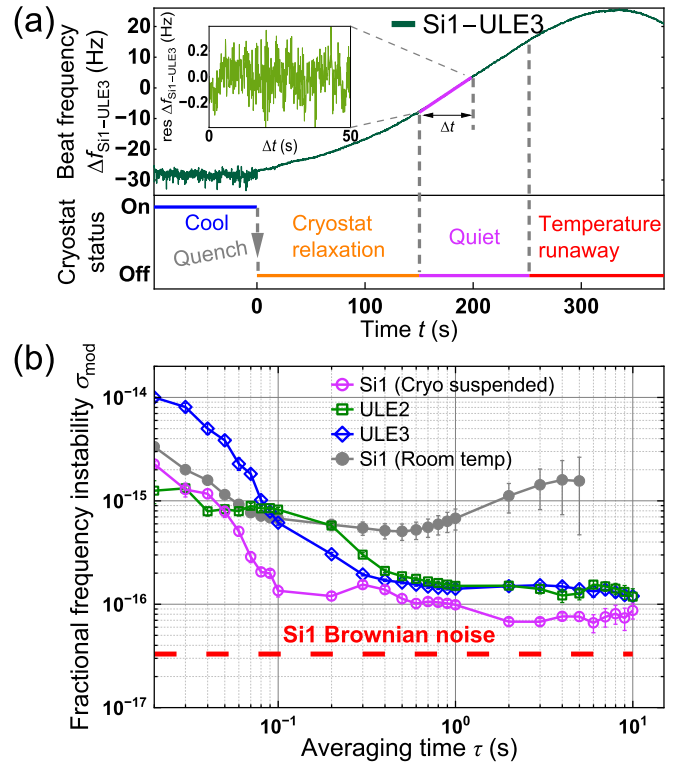


Fig. 2. Dynamically revealing the 10^{-17} -level frequency stability of a cryostat-vibration-free Si1 cavity by cool-quiet quench measurement (CQQM). (a) Diagram of the CQQM. Here, $\Delta f_{\text{Si1-ULE3}}$ denotes an optical beat frequency between Laser1 and Laser3 (mixed down to near d.c.), and $\text{res } \Delta f_{\text{Si1-ULE3}}$ in the inset further denotes the residual frequency noise after subtracting an offset and an overall linear drift from the $\Delta f_{\text{Si1-ULE3}}$ data. (b) Frequency stabilities of the Si1, ULE2, and ULE3 cavities. The vertical axis corresponds to the modified Allan deviation. A dashed line shows a typical Brownian thermal noise floor of the Si1 cavity at 4.9 K. Magenta empty circles denote the frequency instability of the Si1 cavity measured by CQQM (illustrated by the 50-s-long data set in the inset of (a)), which bypasses the cryostat vibration and reveals a sub-total effect mainly contributed by the optical performance of the Si1 cavity and electro-optically induced technical noise sources. Here, improvements (1) to (4) are implemented, while (5) and (6) are not (Table 1). Olive squares and blue diamonds denote the corresponding frequency stabilities of the ULE2 and ULE3 cavities respectively. For comparison, gray solid circles show the substantially larger frequency instability of the Si1 cavity under a room temperature of 292 K. Error bars represent 1σ statistical uncertainties.

rigidity in the whole assembly of the cavity and its supporting structures. Below we depict two rounds of Si1 stability enhancement after addressing these obstacles through the implementations of improvements (1) to (5).

In the first round, we eliminated mirror contamination and improved the rigidity of a vertical supporting structure for the cavity through improvements (1) to (4). Accordingly, the frequency instability of the silicon cavity (denoted as “Si1a”) reached $3.6(3) \times 10^{-16}$ at $\tau = 1$ s and $7.6(8) \times 10^{-17}$ at around 7 s, representing a significant enhancement with respect to the 10^{-15} -level Si0 instability; see Fig. 3a. We further observed the distinct frequency noise power spectral densities (PSDs) of Si1a and Si0. As shown in Fig. 3b, Si0 exhibited an apparent flicker noise at Fourier frequencies of 0.1 to several hertz, which is difficult to model and primarily related to uncontrolled optical scattering with contaminated mirrors. With new mirrors, Si1a showed a much lower, rather flat broadband noise background in its PSD over Fourier frequencies of 0.06 to 5 Hz, which is nevertheless still larger than the reference PSD of Si1-CQQM. The Si1a-versus-Si1-CQQM differential measurements pointed to cryostat vibration as the major limiting factor for Si1a.

In the second round, we further addressed the issue of cryostat vibration by sufficiently increasing the overall rigidity of the whole

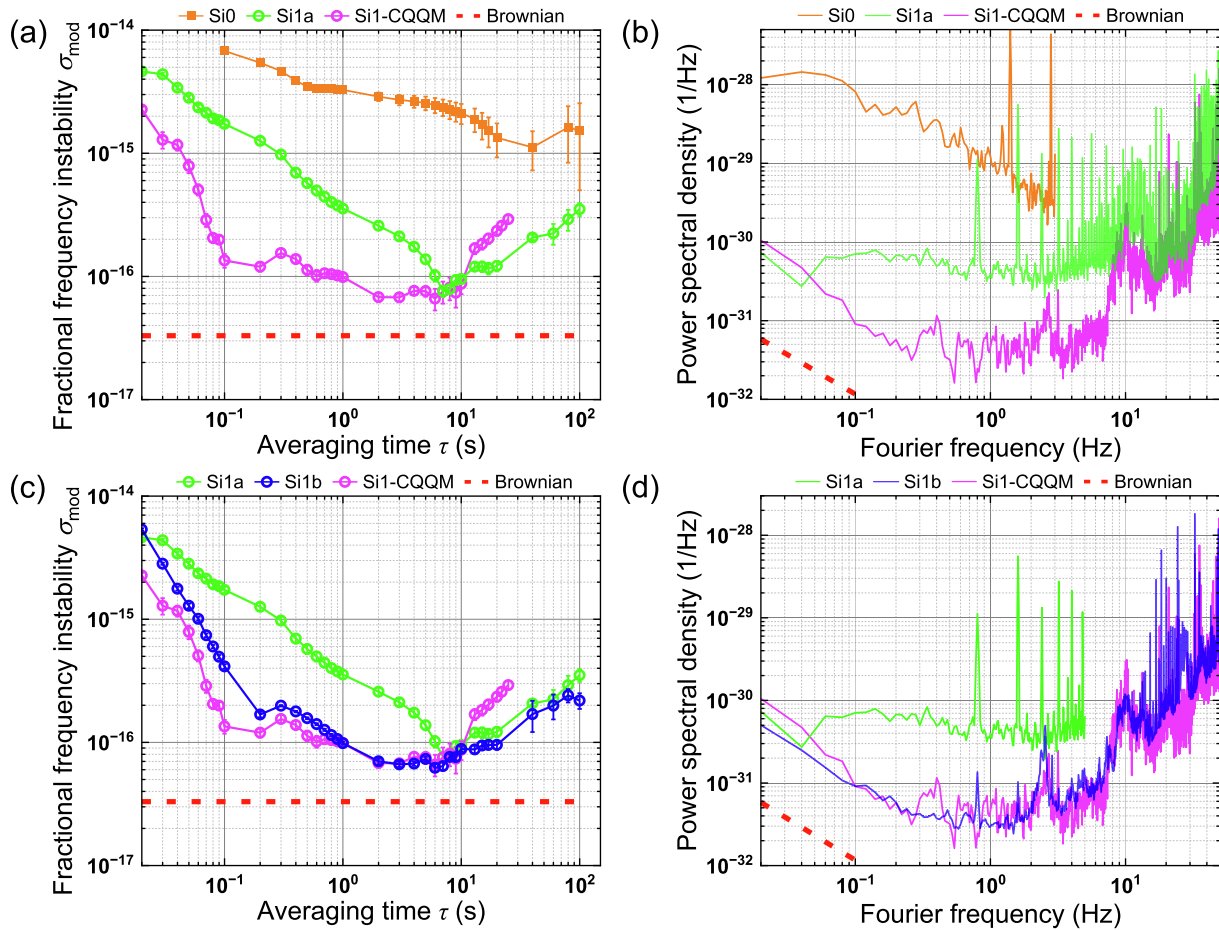


Fig. 3. Overcoming two major obstacles to realizing 10^{-17} -level frequency stability under continuous cooling. (a) Modified Allan deviations. With technical improvements (1) to (4) implemented, the frequency stability of the silicon cavity (“Si1a” configuration, green empty circles) has significantly improved with respect to that of the earlier Si0 cavity with mirror contamination (orange solid squares). (b) Power spectral densities (PSDs) for measurements in (a). (c) Modified Allan deviations. With technical improvements (1) to (5) implemented, the frequency stability of the silicon cavity (“Si1b” configuration, blue empty circles) further shows stability enhancement over Si1a (by a factor of more than three at $\tau = 1$ s). (d) PSDs for measurements in (c). To better show the Si1b PSD, the Si1a PSD is only plotted in part in (d). In (a) to (d), the Brownian thermal noise floor is shown as red dashed lines, and the CQQM results, as a reference for comparison, are shown as magenta empty circles (modified Allan deviations) and magenta lines (PSDs). Error bars represent 1σ statistical uncertainties.

cavity-and-support assembly through improvement (5) (illustrated in Fig. 1a). With improvements (1) to (5) all implemented, we observed an overall stability enhancement for the silicon cavity (denoted as Si1b) with respect to Si1a. The Si1b instability reached $9.8(3) \times 10^{-17}$ at $\tau = 1$ s and improved to $6.2(4) \times 10^{-17}$ at 6 s (Fig. 3c). Remarkably, at Fourier frequencies below 20 Hz, the Si1b PSD reached a level very similar to that of the reference PSD of Si1-CQQM (Fig. 3d). From the differential PSD of Si1b versus Si1-CQQM, we estimated a vastly suppressed residual cryostat-vibration effect that corresponds to a modified Allan deviation below 2×10^{-17} for $\tau \geq 3$ s (see the Supplementary material). This second enhancement can be qualitatively understood as follows. For relatively long cryogenic cavities like Si1, in the absence of a horizontal supporting or constraining structure, the frequency stability becomes susceptible to cryostat-induced *uncontrolled* relative horizontal displacements of the cavity with respect to the platform, which corresponds to an *amplification* of the vibrational effect at the relatively lightweight cavity in comparison to the vibration level at the relatively heavy platform. By implementing improvement (5), we not only increased the effective weight of the cavity by making it form a whole with its horizontal fastening structure, thus reducing the corresponding vibrational sensitivities (see the Supplementary material), but also directly restricted uncontrolled relative horizontal displacements of the cavity to near zero with respect to the plat-

form. Our approach shares certain features with the rigid-mounting method for realizing transportable ultra-stable cavities operating in non-laboratory environments [35–37] and is analogous to the Mössbauer effect in nuclear physics [38,39]. We emphasize that the key function of improvement (5) is not to improve the “environmental” vibrational stability at the platform, but to suppress the vibrational noise at the cavity down to the “environmental level” at the platform. While the platform may still bear more vibrational noise than a base in a room-temperature, cryostat-free cavity system, the relatively heavy weight of the platform (compared with a lightweight free cavity) and its firm connection to the whole sample chamber already make the current “environmental” vibration instability sufficiently small to allow for 10^{-17} -level frequency stabilities of the Si1 cavity at second-scale averaging times. Moreover, numerical simulations showed that, in the presence of a horizontal fastening structure, the effective CTE of the Si1 cavity remains dominated by single-crystalline silicon (see the Supplementary material).

3.4. Optimized Si1 frequency instability reaching 4×10^{-17} under continuous cooling

To optimize the frequency stability of the Si1 cavity, we protected the whole Si1 optical path against acoustic noise, air flow, temperature drift, and other slow fluctuations by implementing

improvement (6). We then performed TCH measurements and analyzed the modified Allan deviations using 27 data sets that were 100 s long for $\tau < 3$ s and 28 data sets that were 500 s long for $\tau \geq 3$ s. As shown in Fig. 4a, the silicon cavity (denoted as Si1c, with improvements (1) to (6) all implemented) demonstrated 10^{-17} -level frequency instability at $\tau = 0.9$ to 100 s, showing advantages of cryogenic silicon cavities over room-temperature cavities ULE2 and ULE3. In particular, the Si1c instability reached $6.0(3) \times 10^{-17}$ at $\tau = 1$ s and improved to $3.4(3) \times 10^{-17}$ at around 6 s. In the range of $\tau = 4$ to 12 s, we extracted an average frequency instability of $4.3(2) \times 10^{-17}$, which is close to the Si1 Brownian thermal noise floor at 4.9 K. At present, the one-second Si1c instability is primarily limited by residual amplitude modulation (RAM) noise (gray dashed line in Fig. 4a). At short and long averaging times, the Si1c instability is dominated by seismic vibration ($\tau < 0.1$ s) and cavity temperature fluctuation ($\tau > 40$ s) respectively; see the Supplementary material. Furthermore, by reducing the platform temperature from 4.9 to 3.3 K, we observed improved long-term instabilities of $4.9(8) \times 10^{-17}$ at $\tau = 100$ s and $5.5(1.2) \times 10^{-16}$ at $\tau = 1000$ s, showing the benefit of reducing the silicon CTE; see Fig. 4a inset and the Supplementary material. In accordance with the modified Allan deviation for 4.9 K, the Si1c PSD (averaged over 140 data sets

that were 100 s long) approached the Si1 Brownian thermal noise floor at Fourier frequencies of 0.04 to 0.06 Hz, showing a marked improvement over Si1b; see Fig. 4b.

We investigated the Si1c ultra-narrow linewidth using two methods. In the first method, based on a systematic scheme demonstrated in Ref. [40] and the Si1c PSDs (140 data sets that were 100 s long), we numerically computed the minimum observable Laser1 linewidth to be $\Delta\nu_{\text{Si1,I}} = 9.6(3)$ mHz (median value) for a measurement time up to 100 s; see Fig. 4c and the Supplementary material. In the second method, we measured the heterodyne beat between Si1 and ULE3 (mixed down to near d.c.) with a digital oscilloscope, as shown by a sample time trace in Fig. 4d. Twelve such time traces of the heterodyne beat were each analyzed via fast Fourier transform and then averaged to yield an average line profile with a fitted full width at half-maximum of $\Delta_{1,3} = 21(1)$ mHz; see Fig. 4e and the Supplementary material. Considering the typical frequency stabilities (averaged over $1 \leq \tau \leq 20$ s) of Si1c and ULE3, $\bar{\sigma}_{\text{Si1c}} \approx 4.5 \times 10^{-17}$ and $\bar{\sigma}_{\text{ULE3}} \approx 1.6 \times 10^{-16}$ respectively, we followed an empirical approach where the linewidths of two stable lasers were added in quadrature in a beat measurement [12], and extracted the Si1 contribution to the beat linewidth as $\Delta\nu_{\text{Si1,II}} = \Delta_{1,3} \bar{\sigma}_{\text{Si1c}} / \sqrt{\bar{\sigma}_{\text{Si1c}}^2 + \bar{\sigma}_{\text{ULE3}}^2} \approx 5.7(3)$ mHz, which is qualitatively consistent with $\Delta\nu_{\text{Si1,I}}$. The ultra-narrow Si1

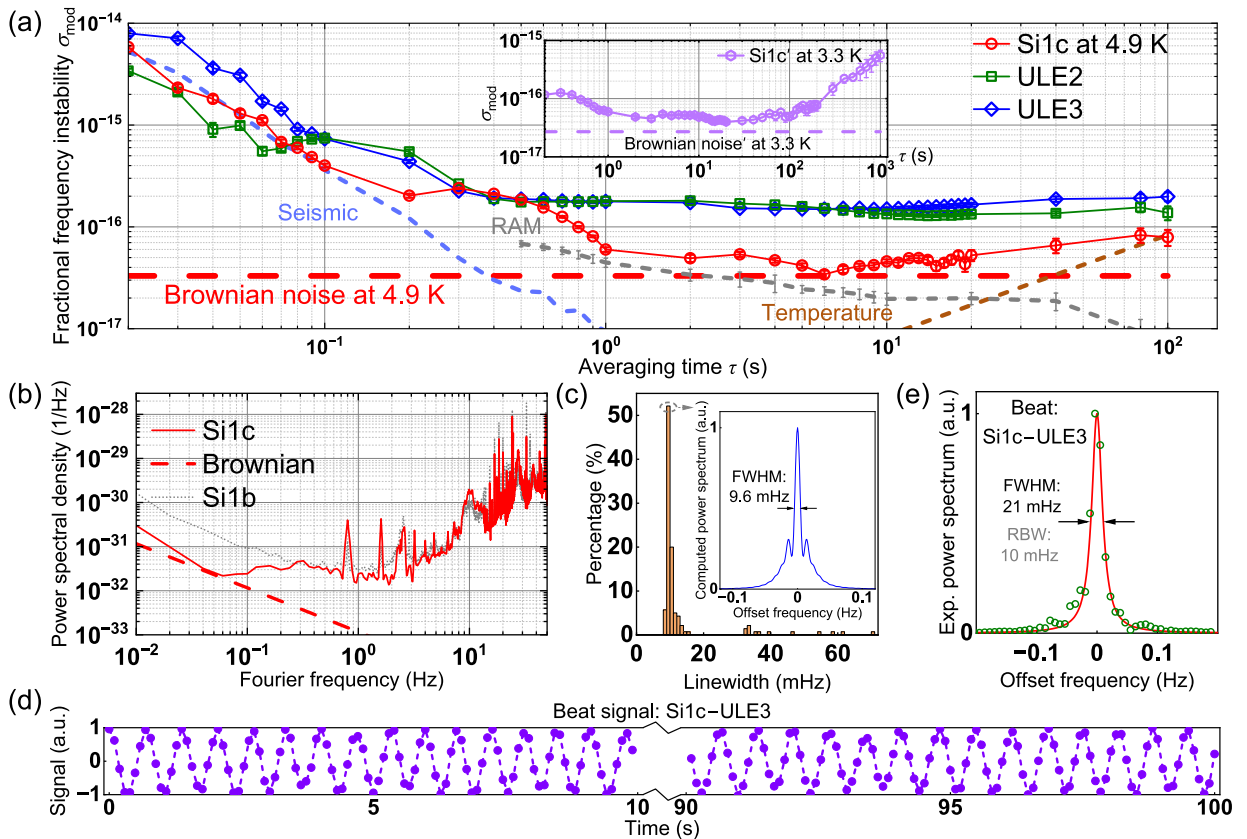


Fig. 4. Optimized frequency instability approaching the Si1 thermal noise floor at 4.9 K under continuous cooling. (a) Modified Allan deviations. Measured frequency instabilities are shown for the silicon cavity with improvements (1) to (6) implemented (“Si1c”, red circles) as well as for ULE2 (olive squares) and ULE3 (blue diamonds) cavities. Red, blue, gray, and brown dashed lines show the computed Brownian thermal noise floor at 4.9 K (as also shown in Figs. 2 and 3), the estimated instability induced by seismic vibrations, the measured instability due to residual amplitude modulation (RAM), and the projected instability due to temperature fluctuation. The inset shows the Si1 frequency instability (purple hexagons) in a configuration (denoted as Si1c’) that is almost the same as Si1c except for a reduced platform temperature of 3.3 K. (b) PSDs. The Si1c spectrum (solid line) is compared with that of Si1b (short dotted line) and the Brownian thermal noise floor. (c) Based on the Si1c PSDs (measured for (b)) and a systematic scheme in Ref. [40], the Si1c linewidths are computed and the histogram constructed, with a median full width at half-maximum (FWHM) of 9.6(3) mHz (shown in the inset of (c)). (d) A sample 100-s-long beat signal (mixed down to near d.c.) and measured with a digital oscilloscope between the Si1 and ULE3 cavities. (e) Average line profile (green circles) for 12 data sets similar to the beat signal measured in (d), analyzed via normalized fast Fourier transform using a 10-mHz resolution bandwidth (RBW) and Hanning window. The profile is fitted to a Lorentzian form (red line). Error bars represent 1σ statistical uncertainties.

linewidth corresponds to an optical quality factor of $\nu_{\text{Si1}}/\Delta\nu_{\text{Si1,l}} \approx 2.2 \times 10^{16}$ for a highly coherent optical local oscillator at 1397 nm, with $\nu_{\text{Si1}} \approx 214.6$ THz.

4. Conclusion and discussion

In summary, we have realized an ultra-stable 10-cm-long silicon cavity at sub-5-K temperatures and demonstrated $4.3(2) \times 10^{-17}$ laser frequency instability that approaches the cavity's Brownian thermal noise floor at 4.9 K. Our work paves the way for enhancing the frequency stabilities of silicon cavities at several-kelvin temperatures using long cavities. With state-of-the-art ultra-low-vibration cryostat techniques [41] and advanced electro-optical control [42], the realization of frequency stability at the 1×10^{-17} or 10^{-18} level is within reach based on a silicon cavity with a length of 20 cm or more. Such highly stable optical local oscillators hold promise for enabling ground- and space-based optical atomic clocks with 10^{-19} to 10^{-21} -level stabilities and other precision interferometric systems, which in turn will provide ultra-sensitive quantum sensors for fundamental science including gravitational waves and dark matter [43–45]. The CQQM method also provides an efficient way to investigate fundamental and electro-optical frequency noise for various new mirror coatings.

Conflict of interest

The authors declare that they have no conflict of interest.

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Author contributions

Xibo Zhang conceived the project. Zhi-Ang Chen, Hao-Ran Zeng, Wen-Wei Wang, and Run-Qi Lei performed the experiments. Zhi-Ang Chen, Wen-Wei Wang, and Han Zhang performed numerical simulations for designing the silicon cavity and associated apparatus. Jian-Zhang Li and Cai-Yin Pang performed the manufacturing and cleansing of the spacer, as well as the optical contacting of the silicon mirrors to the spacer. She-Song Huang contributed important insights for suppressing optical contamination based on high-vacuum environment and for upgrading the cavity support structure to improve the overall rigidity. All authors contributed to the data analysis and the writing of this manuscript.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scib.2025.08.050>.

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